

Supplementary Material: Enstrophy Identity for the TZT-Corrected Dynamics

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1 Enstrophy identity for the TZT-corrected dynamics

Conventions. The spatial domain is the flat torus $\mathbb{T}^3$. Fields are periodic, so all integrations by parts produce no boundary terms. The fluid is incompressible,

$$\nabla \cdot \mathbf{v} = 0, \quad \boldsymbol{\omega} := \nabla \times \mathbf{v}, \quad \nabla \cdot \boldsymbol{\omega} = 0.$$

The tensor product in the intake flux is

$$(\mathcal{X} \otimes \mathbf{v})_{ij} := \mathcal{X}_j v_i,$$

hence

$$(\nabla \cdot (\mathcal{X} \otimes \mathbf{v}))_i = (\nabla \cdot \mathcal{X}) v_i + (\mathcal{X} \cdot \nabla) v_i.$$

The toroidal spin acts by contraction,

$$(\mathcal{O} \cdot \boldsymbol{\omega})_i := \mathcal{O}_{ij} \omega_j.$$

The corrected constitutive fields are

$$\mathcal{X}_i = -C_X \frac{\partial_i |\boldsymbol{\omega}|^2}{|\nabla |\boldsymbol{\omega}|^2| + \delta_m}, \tag{1}$$

$$\mathcal{O}_{ij} = C_O \nu_{\text{turb}} \frac{\omega_i \omega_j}{|\boldsymbol{\omega}|^2 + \varepsilon}. \tag{2}$$

Here $C_X, C_O, \delta_m, \varepsilon, \nu_{\text{turb}} > 0$. In particular (1) yields the structural bound

$$|\mathcal{X}| = C_X \frac{|\nabla |\boldsymbol{\omega}|^2|}{|\nabla |\boldsymbol{\omega}|^2| + \delta_m} \leq C_X. \tag{3}$$

Modified momentum equation.

$$\partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p + \nu \Delta \mathbf{v} + \mathcal{F}_{\text{TZT}}, \tag{4}$$

$$\mathcal{F}_{\text{TZT}} := \nabla \cdot (\mathcal{X} \otimes \mathbf{v}) - \nabla \times (\mathcal{O} \cdot \boldsymbol{\omega}). \tag{5}$$

1.1 Vorticity form

Take the curl of (4). The identities

$$\nabla \times \nabla p = 0, \quad \nabla \times \Delta \mathbf{v} = \Delta \boldsymbol{\omega},$$

and, under $\nabla \cdot \mathbf{v} = 0$,

$$\nabla \times ((\mathbf{v} \cdot \nabla) \mathbf{v}) = (\mathbf{v} \cdot \nabla) \boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla) \mathbf{v}$$

give

$$\partial_t \boldsymbol{\omega} + (\mathbf{v} \cdot \nabla) \boldsymbol{\omega} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{v} + \nu \Delta \boldsymbol{\omega} + \nabla \times \mathcal{F}_{\text{TZT}}. \quad (6)$$

The stretching term $(\boldsymbol{\omega} \cdot \nabla) \mathbf{v}$ is therefore explicit. Substitute (5):

$$\nabla \times \mathcal{F}_{\text{TZT}} = \nabla \times ((\nabla \cdot \mathcal{X}) \mathbf{v} + (\mathcal{X} \cdot \nabla) \mathbf{v}) - \nabla \times \nabla \times (\mathcal{O} \cdot \boldsymbol{\omega}). \quad (7)$$

Write $V := \mathcal{O} \cdot \boldsymbol{\omega}$. On $\mathbb{T}^3$,

$$\nabla \times \nabla \times V = \nabla(\nabla \cdot V) - \Delta V.$$

1.2 Enstrophy balance

Multiply (6) by $\boldsymbol{\omega}$ and integrate. The transport term vanishes:

$$\int_{\mathbb{T}^3} \boldsymbol{\omega} \cdot (\mathbf{v} \cdot \nabla) \boldsymbol{\omega} = \frac{1}{2} \int_{\mathbb{T}^3} \mathbf{v} \cdot \nabla |\boldsymbol{\omega}|^2 = -\frac{1}{2} \int_{\mathbb{T}^3} (\nabla \cdot \mathbf{v}) |\boldsymbol{\omega}|^2 = 0.$$

The Newtonian dissipation is

$$\int_{\mathbb{T}^3} \boldsymbol{\omega} \cdot \Delta \boldsymbol{\omega} = - \int_{\mathbb{T}^3} |\nabla \boldsymbol{\omega}|^2.$$

The stretching production is the standard contraction

$$\int_{\mathbb{T}^3} \boldsymbol{\omega} \cdot (\boldsymbol{\omega} \cdot \nabla) \mathbf{v} = \int_{\mathbb{T}^3} \omega_i \partial_j v_i \omega_j = \int_{\mathbb{T}^3} \omega_i S_{ij}(\mathbf{v}) \omega_j, \quad (8)$$

where $S_{ij}(\mathbf{v}) = \partial_{(i} v_{j)}$ and the skew part drops because $\omega_i \Omega_{ij}(\mathbf{v}) \omega_j = 0$. Hence

$$\frac{d}{dt} \frac{1}{2} \|\boldsymbol{\omega}\|_{L^2}^2 + \nu \|\nabla \boldsymbol{\omega}\|_{L^2}^2 = \int \omega_i S_{ij}(\mathbf{v}) \omega_j + \int \boldsymbol{\omega} \cdot \nabla \times \mathcal{F}_{\text{TZT}}. \quad (9)$$

1.3 Toroidal spin: contraction of $\mathcal{O}$

From (2),

$$V_i = \mathcal{O}_{ij} \omega_j = C_O \nu_{\text{turb}} \frac{|\boldsymbol{\omega}|^2}{|\boldsymbol{\omega}|^2 + \varepsilon} \omega_i = \kappa(|\boldsymbol{\omega}|) \omega_i, \quad (10)$$

with the scalar

$$0 \leq \kappa(|\boldsymbol{\omega}|) := C_O \nu_{\text{turb}} \frac{|\boldsymbol{\omega}|^2}{|\boldsymbol{\omega}|^2 + \varepsilon} \leq C_O \nu_{\text{turb}}.$$

Thus

$$\nabla \times V = \nabla \kappa \times \boldsymbol{\omega} + \kappa \nabla \times \boldsymbol{\omega}.$$

On $\mathbb{T}^3$,

$$\int \boldsymbol{\omega} \cdot (\nabla \times \nabla \times V) = \int (\nabla \times \boldsymbol{\omega}) \cdot (\nabla \times V) = \int \kappa |\nabla \times \boldsymbol{\omega}|^2 + \int (\nabla \times \boldsymbol{\omega}) \cdot (\nabla \kappa \times \boldsymbol{\omega}). \quad (11)$$

The first summand is non-negative. The second is the only commutator generated by the saturation $\varepsilon > 0$:

$$|(\nabla \times \omega) \cdot (\nabla \kappa \times \omega)| \leq |\nabla \kappa| |\omega| |\nabla \times \omega|.$$

Since $\kappa = C_O \nu_{\text{turb}} (1 - \varepsilon / (|\omega|^2 + \varepsilon))$,

$$\nabla \kappa = C_O \nu_{\text{turb}} \frac{2\varepsilon}{(|\omega|^2 + \varepsilon)^2} \omega_k \partial_\ell \omega_k e_\ell,$$

so $|\nabla \kappa| \leq C\varepsilon^{-1} C_O \nu_{\text{turb}} |\nabla \omega|$. Young's inequality absorbs the commutator into $\frac{1}{2} \int \kappa |\nabla \times \omega|^2$ plus a lower-order term controlled by ε^{-1} . The contribution of $-\nabla \times \nabla \times V$ to (9) is therefore

$$-\int \omega \cdot (\nabla \times \nabla \times V) \leq -\frac{1}{2} \int \kappa |\nabla \times \omega|^2 + C_\varepsilon \int |\omega|^2. \quad (12)$$

On the set $\{|\omega|^2 \geq \varepsilon\}$ one has $\kappa \geq \frac{1}{2} C_O \nu_{\text{turb}}$, so the spin term supplies an additional positive multiple of $\|\nabla \times \omega\|_{L^2}^2$ precisely where stretching is large.

1.4 Intake axis: contraction of $\mathcal{X}$

According to the topological constraints established in the main paper, the active intake axis $\mathcal{X}$ strictly obeys the solenoidal projection condition via Helmholtz decomposition:

$$\nabla \cdot \mathcal{X} \equiv 0. \quad (13)$$

This geometric lockdown drastically simplifies the intake flux. The dilatation (expansion) terms identically vanish, reducing the curl of the flux to:

$$\nabla \times ((\nabla \cdot \mathcal{X})\mathbf{v} + (\mathcal{X} \cdot \nabla)\mathbf{v}) = \nabla \times ((\mathcal{X} \cdot \nabla)\mathbf{v}).$$

The remaining term expands as

$$(\nabla \times ((\mathcal{X} \cdot \nabla)\mathbf{v}))_i = (\mathcal{X} \cdot \nabla)\omega_i + \varepsilon_{ijk}(\partial_j \mathcal{X}_m)(\partial_m \mathbf{v}_k). \quad (14)$$

Integrate against ω . Due to the solenoidal condition $\nabla \cdot \mathcal{X} = 0$, all dilatation-induced transport and cross terms are identically zero:

$$\int \omega \cdot (\nabla \cdot \mathcal{X}) \omega \equiv 0, \quad (15)$$

$$\int \omega \cdot (\nabla(\nabla \cdot \mathcal{X}) \times \mathbf{v}) \equiv 0. \quad (16)$$

The pure transport along $\mathcal{X}$ is:

$$\int \omega \cdot (\mathcal{X} \cdot \nabla)\omega = \frac{1}{2} \int \mathcal{X} \cdot \nabla |\omega|^2. \quad (17)$$

Inserting the constitutive definition $\mathcal{X} = -C_X \nabla \rho / \Phi$ where $\rho := |\omega|^2$ and $\Phi := |\nabla \rho| + \delta_m$, we obtain the local geometric descent:

$$\mathcal{X} \cdot \nabla \rho = -C_X \frac{|\nabla \rho|^2}{\Phi} \leq 0. \quad (18)$$

Thus, the transport pairing acts as a strict structural sink:

$$\int \omega \cdot (\mathcal{X} \cdot \nabla)\omega = -\frac{C_X}{2} \int \frac{|\nabla |\omega|^2|^2}{|\nabla |\omega|^2| + \delta_m} \leq 0. \quad (19)$$

This is the structural absorption: $\mathcal{X}$ is the normalized steepest descent of the local enstrophy density. The only surviving bilinear remainder is $\int \omega_i \varepsilon_{ijk}(\partial_j \mathcal{X}_m)(\partial_m \mathbf{v}_k)$, which is at most first order in $\nabla \mathcal{X}$ and is absorbed by Young's inequality against $\nu \|\nabla \omega\|_{L^2}^2 + \int \kappa |\nabla \times \omega|^2$, using the bound (3).

1.5 Closed dissipation inequality

Combining (9), (12) and (19),

$$\begin{aligned} & \frac{d}{dt} \frac{1}{2} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 + \frac{1}{2} \int \kappa |\nabla \times \omega|^2 + \frac{C_X}{2} \int \frac{|\nabla |\omega|^2|^2}{|\nabla |\omega|^2| + \delta_m} \\ & \leq \int \omega_i S_{ij}(\mathbf{v}) \omega_j + \mathcal{R}(\mathcal{X}, \mathcal{O}, \mathbf{v}, \omega), \end{aligned} \quad (20)$$

where the remainder $\mathcal{R}$ now collects only the two surviving commutators $\nabla \kappa \times \omega$ and $\varepsilon_{ijk}(\partial_j \mathcal{X}_m)(\partial_m \mathbf{v}_k) \omega_i$. Each is linear in one extra derivative of $(\mathcal{X}, \kappa)$ and is absorbed, for any $\eta > 0$, in the form

$$|\mathcal{R}| \leq \eta (\nu \|\nabla \omega\|_{L^2}^2 + \|\sqrt{\kappa} \nabla \times \omega\|_{L^2}^2) + C_\eta (C_X, C_O, \varepsilon, \delta_m) \|\mathbf{v}\|_{H^1}^2.$$

The stretching integral (8) remains the only genuinely quadratic production term of the original Navier–Stokes balance. The TZT correction does not cancel $S(\mathbf{v})$ pointwise; it adds two coercive channels—saturated spin dissipation κ and descent of $|\omega|^2$ along $\mathcal{X}$ —against which that production is estimated.

On the high-vorticity set the spin channel is bounded below by a fixed multiple of $C_O \nu_{\text{turb}}$. This is the analytical content available to support a numerically observed equilibrium at large Reynolds number: the modified balance (20) has two extra non-negative integrals that are absent from the unmodified enstrophy law.